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Dynamic device persona identification in a network

Abstract

A network management system (NMS) that can provision and manage a network is provided. During operation, the NMS can determine, for a respective network device in the network, connectivity information indicating how the network device is connected within the network and a device profile of the network device. The NMS can then determine a connectivity tier of the network device in the network based on the connectivity information. The connectivity tier can correspond to a placement of the network device in a hierarchy of network devices in the network. The NMS can select a device persona associated with the connectivity tier. Here, the device persona indicates specification and configurations for the network device. The NMS can send, to the network device, configuration information that configures the device persona on the network device.

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Background/Summary

BACKGROUND

(1) A network device, such as a switch, in a network may support different protocols and services. Furthermore, the network device may be placed at different tiers in a hierarchy of devices in the network. Consequently, the protocols and services supported by individual network devices in the network may vary.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 illustrates an example of a dynamic device persona identification (DDPI) system discovering respective personas of network devices in a network, in accordance with an aspect of the present application.
- (2) FIG. 2 illustrates an example of a DDPI system determining respective connectivity tiers of network devices in a network, in accordance with an aspect of the present application.
- (3) FIG. 3 illustrates an example of a user interface of a DDPI system for enhancing the connectivity tier selection of network devices in a network, in accordance with an aspect of the present application.
- (4) FIG. 4A presents a flowchart illustrating an example of a process of a network management system (NMS) dynamically identifying device persona in a network, in accordance with an aspect of the present application.
- (5) FIG. 4B presents a flowchart illustrating an example of a process of an NMS determining respective connectivity tiers of network devices in a network, in accordance with an aspect of the present application.
- (6) FIG. 5 presents a flowchart illustrating an example of a process of an NMS enhancing the connectivity tier selection of network devices in a network, in accordance with an aspect of the present application.
- (7) FIG. 6 illustrates an example of a computing system facilitating DDPI in a network, in accordance with an aspect of the present application.
- (8) FIG. 7 illustrates an example of a non-transitory computer-readable storage medium that facilitates DDPI in a network, in accordance with an aspect of the present application.
- (9) In the figures, like reference numerals refer to the same figure elements.

DETAILED DESCRIPTION

(10) A network can facilitate data exchange for a variety of applications running on physical and virtual devices. Such applications have brought a progressively increasing amount of network traffic. As a result, network devices, such as switches, are often built with versatile capabilities. To do so, a network device may support different protocols and services. Furthermore, the network device can participate in different types of networks while operating in different roles. For example, the network device can operate as a participating switch in a distributed virtual switch

(DVS) where a plurality of participating switches can operate in conjunction with each other to facilitate the operations of the DVS. While operating as a participating switch in a DVS, the network device can also operate as an access switch that connects end devices (or user devices) to the network.

(11) Since each network device in the network may support different types of operations for different roles, providing automated configuration to the network devices can include determining corresponding personas for the network devices. The persona can be a “template” set of configurations applicable to the same category of network devices (i.e., the network devices that are expected to behave in the same manner). Determining a persona for individual network devices may require extensive analysis. Traditionally, such an analysis is performed by a user, such as an administrator. For example, the administrator can configure each network device of the network based on the topology (e.g., how the network device is connected in the network) and the persona of the network device. Therefore, facilitating automatic configurations to a large network with a significant number of network devices can be challenging.

(12) The aspects described herein address the problem of dynamically identifying the persona for a network device in a network by (i) determining how the network devices are connected to each other in the network; (ii) classifying the network device to a connectivity tier based on a number of connections via the network device; and (iii) selecting a persona corresponding to the connectivity tier for the network device. The connectivity tier can indicate the placement of the network device in a hierarchy of network devices in the network (e.g., at the access, aggregation, or core tier). By matching the connectivity tier of the network device to a corresponding persona, the network device can be dynamically allocated with the persona that allows it to efficiently operate at the connectivity tier.

(13) With existing technologies, to deploy a network, the administrator can connect the network devices with each other in the network. Based on the connections, the network devices can logically couple to a network management system (NMS), such as a network orchestrator, and initiate communication with the NMS. A logical connection can span a number of nodes, links, and networks (e.g., via the cloud). The NMS can provision and manage the network devices of the network using the corresponding logical connections. Being logically coupled to the NMS for provisioning and management can be referred to as “onboarding.”

(14) The administrator can then configure, via the NMS, each network device based on how it is connected in the network and the persona of the network device. Hence, the deployment process can rely on the administrator's knowledge of the persona of each network device. Subsequently, the administrator may identify the topology of the network and configure individual network devices based on the persona of the network device. When the network is complex with different types of network devices (e.g., access and aggregation switches), configuring individual network devices based can be inefficient, cumbersome, and error prone.

(15) To address this problem, the NMS of the network can deploy a DDPI system capable of dynamically identifying the persona of a respective network device in the network. During operation, when a network device onboards with the NMS, the network device can send a discovery packet to the NMS. The discovery packets can be Link Layer Discovery Protocol (LLDP) change notification packets. A respective discovery packet can include a set of discovery information associated with a corresponding network device. The discovery information can include one or more of: a numeric port description, virtual local area network (VLAN) information associated with a respective port, an alpha-numeric name of the network device, an alpha-numeric system description (e.g., a description of the network device's hardware and operating system), system capabilities indicating the layer-2 and layer-3 services supported by the network device, one or more network addresses (e.g., media access control (MAC) and Internet Protocol (IP) addresses) allocated to the network device, and a list of protocols operating on the network device.

(16) When the NMS receives the discovery packet from the network device, the DDPI system can

obtain the discovery information and store the information in a database. In some examples, the database can be a graph database, such as ArangoDB. A graph database can be a specialized database that uses edges and nodes of a graph structure for storing data and applying semantic queries to the data. The data in the graph database can be stored based on the Internet Engineering Task Force (IETF) draft on JavaScript Object Notation (JSON) schema. Because the graph database can be used to represent the network based on edges and nodes, the DDPI can use the graph database to classify the network device.

(17) Upon receiving the discovery packets from the network devices of the network, the DDPI system can apply a classification process on the discovery information. The DDPI system can determine the connectivity information from the discovery information. The connectivity information can indicate how each network device is connected to other network devices in the network. Based on the connectivity information of the network, the DDPI system can explore the paths in the network. Exploring the paths can include, from each network device, traversing a path from each connected port of the network device up to a predetermined number of hops. Each path traversal can include exploring a subsequent network device reachable via a link at each hop. The DDPI system can repeat the path traversal for all network devices.

(18) The DDPI system can then determine the number of paths through a network device as a connectivity number for the network device. Therefore, the connectivity number can indicate the number of connections for the network device. Based on the connectivity number, the DDPI system can determine the connectivity tier of individual network devices. The connectivity tier can indicate the placement of the network device in the hierarchy of network devices in the network. For example, the hierarchy of the network devices can include an access tier coupling end devices, an aggregation tier that aggregates traffic from a plurality of access devices, and a core or border tier that can couple the network to an external network (e.g., to a wide-area network or WAN, such as the Internet).

(19) A network device at the access tier typically can communicate with another network device at any tier via the aggregation tier. Therefore, the network devices in the aggregation tier may have the highest number of connections through them. On the other hand, the network devices in the access tier may have a few connections through them. Accordingly, if the connectivity number for a network device is below a threshold, the DDPI system can determine an access tier as the connectivity tier of the network device. Hence, if the network device is a switch, the DDPI system can identify the network device as an access switch. On the other hand, if the connectivity number for the network device is above the threshold, the DDPI system can determine whether the network device is connected to any device in the access tier (e.g., to an access switch). If such a connection exists, the DDPI system can determine an aggregation tier as the connectivity tier of the network device. Otherwise, the DDPI system can determine a core (or border) tier as the connectivity tier of the network device.

(20) The DDPI system may also perform validations based on the respective device profiles of the network devices. The DDPI system may obtain identifying information associated with a network device from a discovery packet and determine the corresponding device profile. The identifying information can include vendor-specific information, such as a model number. The device profile can include a corresponding model number and device capabilities. The DDPI system can identify the device profile by matching the model number obtained from the discovery packet with the identified device profile.

(21) The DDPI system can then use the information in the device profile to validate the allocation of a connectivity tier. For example, if the model number indicates a high-performance device suitable as a core network device, the DDPI system may not determine an access tier for that network device. In the same way, the DDPI system can utilize the device capabilities (e.g., high-capacity ports, multi-band optical transponders, etc.) to determine whether a network device should be at the core tier. Therefore, in addition to the connectivity to the access tier, the DDPI system can

check against the device profile and device capabilities to determine whether a network device belongs to an aggregation tier or a core tier.

(22) The DDPI system can also determine additional information for a network device. For example, the DDPI system can determine whether the network device has a direct connection to another network device in the same tier. If the network device has such a connection, the DDPI system can determine that the network device is deployed as a device pair (e.g., a DVS). The DDPI system can select a persona in accordance with the deployment. The DDPI system may maintain one or more predefined personas for each connectivity tier. Based on the connectivity tier and additional information associated with a network device, the DDPI system can select a persona for the network device. The DDPI system can then determine the set of configurations indicated by the persona and “push” the configurations to the device. In this way, the DDPI system can automatically identify the persona of the network device and configure the network device accordingly.

(23) In this disclosure, the term “switch” is used in a generic sense, and it can refer to any standalone network device or fabric device operating in any network layer. “Switch” should not be interpreted as limiting examples of the present invention to layer-2 networks. Any device that can forward traffic to an external device or another switch can be referred to as a “switch.” If the switch is a virtual device, the switch can be referred to as a virtual switch.

(24) Furthermore, if a network device facilitates communication between networks, the network device can be referred to as a gateway device. Any physical or virtual device (e.g., a virtual machine or switch operating on a computing device) that can forward traffic to an end device can be referred to as a “network device.” Examples of a “network device” include, but are not limited to, a layer-2 switch, a layer-3 router, a routing switch, a component of a Gen-Z network, or a fabric switch comprising a plurality of similar or heterogeneous smaller physical and/or virtual switches.

(25) The term “packet” refers to a group of bits that can be transported together across a network. “Packet” should not be interpreted as limiting examples of the present invention to a particular layer of a network protocol stack. “Packet” can be replaced by other terminologies referring to a group of bits, such as “message,” “frame,” “cell,” “datagram,” or “transaction.” Furthermore, the term “port” can refer to the port that can receive or transmit data. “Port” can also refer to the hardware, software, and/or firmware logic that can facilitate the operations of that port.

(26) FIG. 1 illustrates an example of a DDPI system discovering respective personas of network devices in a network, in accordance with an aspect of the present application. A network **100** can include a number of network devices, and may include heterogeneous network components, such as layer-2 and layer-3 hops, and tunnels. In some examples, network **100** can be an Ethernet network, InfiniBand network, or other network, and may use a corresponding communication protocol, such as Internet Protocol (IP), FibreChannel over Ethernet (FCoE), or other protocol. Network **100** can include a number of network devices, such as network devices **102**, **104**, and **106**. Network **100** can include other network devices not shown in FIG. 1. A respective network device in network **100** can be associated with a MAC address and an IP address.

(27) In some examples, network **100** can be a distributed tunnel fabric, where the switches can be coupled to each other via tunnels. Network **100** can then also be referred to as fabric **100**. In fabric **100**, tunnel encapsulation is initiated and terminated within fabric **100**. Network devices in fabric **100** may form a mesh of tunnels. Examples of a tunnel can include, but are not limited to, Virtual Extensible Local Area Network (VXLAN), Generic Routing Encapsulation (GRE), Network Virtualization using GRE (NVGRE), Generic Networking Virtualization Encapsulation (Geneve), Internet Protocol Security (IPsec), and Multiprotocol Label Switching (MPLS). The tunnels in fabric **100** can be formed over an underlay network. The underlay network can be a physical network, and a respective link of the underlying network can be a physical link. A respective switch pair in the underlay network can be a border gateway protocol (BGP) peer. The switches in the underlay network can use the BGP to establish routes via which encapsulated packets of fabric **100**

are forwarded in the underlay network. A VPN, such as an EVPN, can be deployed over fabric **100**. (28) In network **100**, two or more network devices can operate as a single virtual switch in conjunction with each other to facilitate a DVS. The DVS may operate as the gateway device in network **100** and facilitate external communication for network **100**. The DVS can also operate as an access device that connects end devices (or user devices) to network **100**. The DVS can be associated with one or more virtual network addresses (e.g., a virtual IP address and/or a virtual MAC address). Consequently, all switches participating in the DVS can be associated with the virtual network addresses. The virtual network addresses allow the participating network devices to facilitate high availability to each other. For example, if one of these network devices becomes unavailable due to a failure or power cycle, the other network device can remain associated with the virtual network addresses and continue to receive and forward packets based on the virtual network addresses.

(29) With existing technologies, to deploy network **100**, a user, such as administrator **120**, can connect the network devices with each other in network **100**. Based on the connections, the network devices of network **100** can onboard with NMS **140**. Administrator **120** can then provision and manage the network devices of network **100** via NMS **140**. For example, to provision network device **102**, administrator **120** can provide a set of configuration parameters (e.g., the VLANs enabled at each port) to network device **120** via a user interface of NMS **140**. Administrator **120** can determine the configuration parameters based on how network device **102** is connected in network **100** and the corresponding persona. Hence, the deployment of network device **102** in network **100** can rely on administrator **120**'s knowledge of the persona of network device **102** and the topology of network **100**. If network **100** is complex with different types of network devices, configuring individual network devices in network **100** based can be inefficient, cumbersome, and error prone.

(30) To address this problem, NMS **140** can deploy a DDPI system **150** that can dynamically identify the persona of the network devices in network **100**. When network devices of network **100** onboard with NMS **140**, they can send corresponding discovery packets **112** to NMS **140** (e.g., via the cloud). Discovery packets **112** can be LLDP change notification packets. A respective packet of discovery packets **112** can include a set of discovery information associated with a corresponding network device. For example, a discovery packet **114** from network device **102** can comprise discovery information associated with network device **102**, which can include one or more of: port descriptions of a respective port of network device **102**, VLAN information associated with the port, an alpha-numeric name of network device **102**, the description of the hardware and operating system of network device **102**, system capabilities indicating the layer-2 and layer-3 services supported by network device **102**, one or more network addresses (e.g., MAC and IP addresses) allocated to network device **102**, and a list of protocols operating on network device **102**.

(31) When NMS **140** receives discovery packet **114** from network device **102**, a data aggregation process **152** of DDPI system **150** can obtain the discovery information from packet **114** and store the information in a discovery database **132** of DDPI system **150**. In some examples, database **132** can be a graph database, such as ArangoDB. In this way, data aggregation process **152** can obtain discovery information **116** from discovery packets **112** from network **100** and store discovery information **116** in database **132**. Discovery information **116** can include port information **122** (e.g., description and VLAN of a respective port), route information **124** (e.g., locally determined routes), and system description and capabilities **126** (e.g., a description of the hardware and operating system, and the supported services) associated with each network device.

(32) DDPI system **150** can apply a classification process **154** on discovery information **116** in database **132**. Classification process **154** can determine the connectivity information from discovery information **116**. The connectivity information can indicate how each network device is connected to other network devices in network **100**. Based on the connectivity information of network **100**, classification process **154** can explore the paths in network **100**. Exploring the paths

can include, from each network device, traversing a path from each connected port of the network device up to a predetermined number of hops, X. The value of X can be configured by administrator **120** via an interface **160** of DDPI system **150**. Examples of interface **160** can include, but are not limited to, a graphical user interface (GUI), a web interface, a textual interface, a touch-based interface, a gesture-based interface, and a movement-based interface.

(33) Classification process **154** can then determine the number of paths through a network device as a connectivity number for the network device. For example, if N paths within X hops pass through network device **102**, the connectivity number of network device **102** can be N. Similarly, if M paths within X hops pass through network device **104**, the connectivity number of network device **104** can be M. Hence, the pass-through rates for network devices **102** and **104** can be N and M, respectively. Accordingly, the connectivity number can indicate the number of connections from a network device. Based on the connectivity number, classification process **154** can determine the connectivity tier of individual network devices in network **100**. The connectivity tier can indicate the placement of a network device in the hierarchy of network devices, such as access, aggregate, and core device, in network **100**.

(34) Typically, in network **100**, the aggregation tier may have the highest number of connections through them. In other words, the aggregation tier may have the highest pass-through rate. Here, the pass-through rate can indicate the number of paths passing through a network device. On the other hand, the network devices in the access tier may have a few connections through them. For example, if the connectivity number, N, for network device **102** is below a threshold, classification process **154** can allocate network device **102** to an access tier. The threshold can be a preconfigured value. Similarly, if the connectivity number for network device **106** is below a threshold, classification process **154** can allocate network device **106** to an access tier. On the other hand, if the connectivity number, M, for network device **104** is above the threshold, classification process **154** can determine whether network device **104** is connected to any device in the access tier. If network device **104** is connected to the access tier (e.g., to network device **102**), classification process **154** can allocate network device **104** to an aggregation tier. Otherwise, classification process **154** can allocate network device **104** to a core (or border) tier.

(35) DDPI system **150** can include an enhancement process **156** that can perform validations based on device profiles **118** of the network devices in network **100**. DDPI system **150** can maintain a device repository **134** (e.g., a relational database) that stores device profiles **118**. Device profiles **118** can include the respective model numbers and capabilities of the network devices. DDPI system **150** may obtain identifying information associated with a network device from a discovery packet. The identifying information can include vendor-specific information, such as a model number. DDPI system **150** can identify the device profile by matching the model number obtained from the discovery packet with the identified device profile from repository **134**.

(36) DDPI system **150** can use the information in the device profile to validate the allocation of a connectivity tier. For example, if the model number corresponds to network device **104** suitable as a core network device, enhancement process **156** may not determine an access tier for network device **104**. In the same way, enhancement process **156** can utilize the device capabilities (e.g., high-capacity ports, multi-band optical transponders, etc.) to determine whether network device **104** should be at the core tier. Therefore, in addition to the connectivity to the access tier, enhancement process **156** can check against device profiles **118** and device capabilities to determine whether network device **104** belongs to an aggregation tier or a core tier.

(37) Classification process **154** can also determine additional information for a network device. For example, classification process **154** can determine whether network device **102** has a direct connection to another network device **106** in the same tier. If network devices **102** and **106** are directly connected, classification process **154** can determine that network devices **102** and **106** are deployed as a device pair (e.g., a DVS). Classification process **154** can select a persona in accordance with the deployment. DDPI system **150** may maintain one or more predefined personas

for each connectivity tier in a persona repository **136** (e.g., a data structure). Based on the connectivity tier and additional information associated with a network device, classification process **154** can select a persona for the network device.

(38) Enhancement process **156** can present user interface **160** that can allow administrator **120** to swap a persona if classification process **154** has misclassified the persona or the recommended persona does not match administrator **120**'s intention. Furthermore, if administrator **120** changes a persona selected by classification process **154**, DDPI system **150** can recalculate the threshold accordingly and enhance classification process **154**. Once a persona is selected for a network device, configuration process **158** of DDPI system **150** can then determine a set of configurations associated with the persona and “push” the configurations to the network device.

(39) For example, once classification process **154** allocates network device **102** as an access device and selects a corresponding persona **142** from repository **136**, configuration process **158** can send the configurations indicated by persona **142** to network device **102**. Similarly, when classification process **154** allocates network device **104** as an aggregate device and selects a corresponding persona **144** from repository **136**, configuration process **158** can send the configurations indicated by persona **144** to network device **104**. In this way, DDPI system **150** can automatically identify the persona of network devices in network **100** and configure the respective personas on the network devices accordingly.

(40) FIG. 2 illustrates an example of a DDPI system determining respective connectivity tiers of network devices in a network, in accordance with an aspect of the present application. A large-scale network **200**, such as an enterprise network, can include a number of network devices. Core or border network devices **202** and **204** can couple network **200** with a WAN **250**, such as the Internet. A number of end devices **220** can be coupled to network **200** via edge or access network devices **212**, **214**, **216**, and **218**. Aggregation network devices **206** and **208** can aggregate traffic from end devices. An NMS **252** reachable via WAN **250** can provision, configure, and manage network **200**. To efficiently identify the respective personas of the network devices of network **200**, a DDPI system **254** can be deployed (e.g., in conjunction with NMS **252**).

(41) In network **200**, traffic to and from end devices **220** can traverse through network devices **206** and **208** for both north-south (e.g., to and from WAN **250**) and east-west (e.g., between end devices) traffic flows. For example, traffic from a network server **260**, such as a web server, to end device **222** can be forwarded via network device **202** or **204**, and subsequently via network device **206** or **208**. On the other hand, traffic from end device **222** to end device **224** is also forwarded via network device **206** or **208**. Consequently, network devices **206** and **208** may have the highest pass-through rates in network **200** followed by the pass-through rates of network devices **202** and **204**. Inversely, network devices **212**, **214**, **216**, and **218** can have relatively lower pass-through rates.

(42) Based on this traffic pattern, DDPI system **254** can determine the respective connectivity numbers of the network devices of network **200**. When the network devices of network **200** onboard with NMS **252**, these network devices can provide their discovery information to NMS **252**. NMS **252** can determine how each network device is connected to other network devices in network **200**. NMS **252** can apply a topology generation service on the discovery information to determine the connectivity information. DDPI system **254** can then explore the paths in network **200** from each network device up to a predetermined number of hops. In the example of FIG. 2, the value of the predetermined number of hops can be five. The value can be configured by administrator **240** via an interface of DDPI system **254** based on the topology and tiering of network **200** and the efficiency of execution. For example, if the paths are longer, DDPI system **254** may traverse more devices and cause inefficient execution.

(43) DDPI system **254** can then determine the number of paths that include a network device as a connectivity number for the network device. For example, if N paths within five hops are traversed through network device **206**, the connectivity number of network device **206** can be N. DDPI system **254** can repeat the exploration of paths from a respective network device in network **200**.

For example, DDPI system **254** can select an initial or source network device, such as network device **202** (denoted with a dashed-line enclosure), and start the exploration from network device **202**. The next hop of network device **202** can be network device **204** or **206**. A traversal path **262** from network device **202** can include network device **206** as the first hop (H=1). Here, H can indicate the hop count from network device **202**. Another traversal path **264** can include network device **204** as the first hop (H=1).

(44) After network device **206**, traversal path **262** can have five possible network devices for the next hop. Among the five possible network devices, DDPI system **254** can select network device **212**. The selection can be based on one or more selection criteria (e.g., round-robin or random selection). Even though traversal path **262** can include network device **212** as the second hop (H=2), DDPI system **254** explores respective paths through all five possible network devices as the second hop. In this way, DDPI system **254** can explore all possible paths from network device **202**. Traversal path **262** can subsequently include network devices **208**, **214**, and **206** (H=3, 4, and 5, respectively) as the subsequent hops until the five-hop limit is reached. In accordance with traversal path **262**, DDPI system **254** can increment the connectivity numbers of network device **202** to one.

(45) On the other hand, after network device **204**, traversal path **264** can include network device **208** (H=2). Traversal path **264** can then have five possible network devices for the next hop, each of which can be explored by DDPI system **254**. Traversal path **264** can subsequently include network devices **218**, **206**, and **216** (H=3, 4, and 5, respectively) as the subsequent hops until the five-hop limit is reached. In accordance with traversal path **264**, DDPI system **254** can increment the connectivity numbers of network device **202** to two. By repeating this process for all possible paths from network device **202**, DDPI system **254** can incorporate the number of connections originating at network device **202** into the connectivity numbers of corresponding network devices. Here, each path can include five edges, and each edge in the path can be labeled with a unique key in a sorted combination of the corresponding device identifiers. For example, the edge between network devices **202** and **206** can be identified by the identifier pairs of network devices **202** and **206**. The identifier can be a network address, such as an IP or a MAC address.

(46) DDPI system **254** can use a bucketing algorithm to classify network devices of network **200** by comparing respective connectivity numbers with a threshold **230**. DDPI system **254** can dynamically determine the value of threshold **230**. Threshold **230** may also be configured by an administrator. To dynamically determine the value of threshold **230**, DDPI system **254** can sum the connectivity numbers of all network devices in network **200** and divide the result of summation by the number of network devices in network **200**. DDPI system **254** may adjust the value of threshold **230** based on an adjustment value. The adjustment value can be a value between zero and one, and may be multiplied to the result of the division.

(47) Using the bucketing algorithm, DDPI system **254** can then identify the network devices with connectivity numbers below threshold **230** and place them in an access bucket **232**. DDPI system **254** can place the rest of the network devices in an upper bucket **234**. Subsequently, DDPI system **254** can classify the network devices in bucket **232** to the access (or edge) tier. Among the network devices in bucket **234**, DDPI system **254** can determine which network devices are connected to at least one network device in bucket **232** (i.e., has an access connection) and classify them to the aggregation tier. DDPI system **254** can then place the rest of the network devices in bucket **234** to the core (or border) tier.

(48) DDPI system **254** can also determine which network devices in network **200** have direct connections within the same tier. For example, DDPI system **254** can determine that network device **206** has a direct connection to another network device **208** in the same tier. Since network devices **206** and **208** are directly connected, DDPI system **254** can determine that network devices **206** and **208** are deployed as a device pair (e.g., a DVS). DDPI system **254** can select a persona for network devices **206** and **208** that can allow them to operate as a high-availability switch pair. In

other words, the persona selected for network devices **206** and **208** can incorporate configurations that allow them to operate in the aggregation tier while operating in a high-availability device pair. Hence, for the same tier, DDPI system **254** may maintain multiple personas and can select the one that matches the tier as well as additional relevant information (e.g., presence in a DVS).

(49) FIG. **3** illustrates an example of a user interface of a DDPI system for enhancing the connectivity tier selection of network devices in a network, in accordance with an aspect of the present application. A DDPI system **354**, in conjunction with an NMS **352**, can present a user interface **300**. Examples of interface **300** can include, but are not limited to, a GUI, a web interface, a textual interface, a touch-based interface, a gesture-based interface, and a movement-based interface. A user **310** (e.g., an administrator) may access user interface **300** via a WAN **350** (e.g., via the Internet). Interface **300** can present a topology discovered by NMS **352** and the corresponding tiers assigned DDPI system **354** to user **310**.

(50) For example, the discovered topology of a network can comprise network devices **302**, **304**, **306**, and **308**. DDPI system **354** can assign network device **302** to an aggregation tier, and network device **304**, **306**, and **308** to an access tier. Interface **300** can allow user **310** to swap the persona (i.e., the allocated tier) if DDPI system **354** has misclassified a network device or the recommended tier does not match user **310**'s intention. For example, if user **310** selects a network device, such as network device **308** (denoted with a dashed enclosure), a tier selection interface **320** (e.g., a pop-up window) can appear on interface **300**. Interface **320** can include the recommended tier preselected and can support the option for user **310** to select another tier. For example, for network device **308**, the access tier can be preselected in interface **320**. User **310** can select another tier, such as the aggregation or core tier, for network device **308** via interface **320**.

(51) FIG. **4A** presents a flowchart illustrating an example of a process of an NMS dynamically identifying device persona in a network, in accordance with an aspect of the present application. During operation, the NMS can provision and manage a network (operation **402**). The provisioning can include configuring the network devices of the network. To facilitate the configurations to the network devices, the NMS can maintain, in a local repository, one or more predefined personas for each connectivity tier (operation **404**). For example, the NMS can maintain respective personas for access, aggregation, and core tiers. Furthermore, at each tier, the NMS may maintain multiple personas.

(52) To classify the network devices, the NMS can determine, for each network device, connectivity information indicating how the network device is connected within the network. The NMS can also determine a device profile of the network device (operation **406**). The device profile can indicate the capabilities of the network device. For example, the device profile can include the model number of the network device based on which the NMS can determine the capabilities. The NMS can determine a connectivity tier of the network device in the network based on the connectivity information (operation **408**). By determining the connectivity tier, the NMS can determine the placement of the network device in the hierarchy of network devices in the network based on the connectivity tier (operation **410**).

(53) The NMS can apply a bucketing algorithm to identify the network devices in the access tier. If the connectivity number of a network device is below a threshold, the NMS can assign the access tier to the network device. The NMS can also identify the network devices in the aggregation and core tiers based on the connectivity to the access tier. In this way, the NMS can automatically determine a connectivity tier. The NMS can then validate the connectivity tier against the device profile, which includes vendor-specific information (e.g., a model number), to determine whether the connectivity tier is correctly allocated (operation **412**). For example, if the NMS determines, based on the device profile, that a network device is a high-capacity device, the network device may not be allocated to the access tier. The NMS can then determine whether the network device is connected within the same connectivity tier (e.g., has a link within the same tier) (operation **414**).

(54) If the network device is connected within the same connectivity tier, the NMS can indicate that

the network device is in a device pair facilitating high availability to each other (operation **416**). The direct connection can be an inter-switch link (ISL) via which a network device can detect the unavailability and facilitate a failover accordingly. If the network device is not connected within the same connectivity tier (operation **414**) or upon indicating the high availability (operation **416**), the NMS can select a device persona associated with the connectivity tier from the repository, where the device persona indicates the specification and configurations for the corresponding network device (operation **418**). The NMS can then send, to the network device, configuration information that configures the device persona on the network device (operation **420**).

(55) FIG. **4B** presents a flowchart illustrating an example of a process of an NMS determining respective connectivity tiers of network devices in a network, in accordance with an aspect of the present application. During operation, the NMS can receive an LLDP packet based on a logical connection with a network device. Here, the logical connection indicates the network device being provisioned and managed by the NMS (operation **452**). The logical connection with the NMS indicates that the network device has been onboarded to the NMS. Accordingly, the NMS can receive the LLDP packet comprising discovery information associated with the network device. The NMS can then obtain the discovery information from the LLDP packet (operation **454**).

(56) The NMS can determine the number of connections associated with the network device based on the discovery information (operation **456**). The NMS can explore the paths within a predetermined number of hops from each network device of the network. The NMS can determine the number of paths passing through the network device as the connectivity number. The NMS can then determine whether the connectivity number is less than a threshold (operation **458**). If the connectivity number is less than a threshold, the NMS can determine an access tier as the connectivity tier of the network device (operation **460**).

(57) If the connectivity number is not less than the threshold (i.e., greater than or equal to the threshold), the NMS can also determine whether the network device is connected to the access tier (operation **462**). The connectivity to the access tier indicates that the network device is to aggregate traffic from the access-tier network devices. Accordingly, if the network device is connected to the access tier, the NMS can determine an aggregation tier as the connectivity tier of the network device (operation **464**). On the other hand, if the network device is not connected to the access tier, the placement of the network device can be beyond the aggregation tier within the hierarchy of the network devices in the network. Hence, the NMS can determine a core tier as the connectivity tier of the network device (operation **466**).

(58) Upon determining the connectivity tier (operation **460**, **464**, or **466**), the NMS can determine the device profile based on the identifying information associated with the network device in the LLDP message (operation **468**). The LLDP packet can include vendor-specific information, such as a model number of the network device. The NMS can match the model number in the LLDP packet with the model number in the device profile and select the device profile for the network device. The NMS can then validate the connectivity tier based on the device profile (operation **470**). For example, if the device indicates a high-capacity network device, the NMS may not allocate the network device to an access tier. The NMS can then send the configuration information to each network device at the access tier prior to sending the configuration information to a network device in the aggregation tier (operation **472**). In this way, the NMS can configure the network device from a bottom-up approach.

(59) FIG. **5** presents a flowchart illustrating an example of a process of an NMS enhancing the connectivity tier selection of network devices in a network, in accordance with an aspect of the present application. During operation, the NMS can present a user interface to indicate the determination of a connectivity tier of a network device (operation **502**). The user interface can present the determined topology of the network and the connectivity tier allocated to each of the network devices. The topology can show the hierarchy of the network devices in the network. The NMS can receive an instruction from a user via the user interface (operation **504**). The instruction

can be an updated selection of a connectivity tier for a network device of the network. The instruction can also be a confirmation of a recommended connectivity tier.

(60) Accordingly, the NMS may determine whether the instruction provides an update (operation **506**). If the instruction provides an update, the NMS can update the connectivity tier of the network device based on the instruction (operation **508**). For example, if the recommended connectivity tier is an aggregation tier, and the instruction updates it to an access tier, the NMS can update the connectivity tier of the network device to the aggregation tier. On the other hand, if the instruction does not provide an update, the instruction can be the confirmation of the recommended connectivity tiers. The NMS can then confirm the connectivity tier of the network device (operation **510**).

(61) FIG. **6** illustrates an example of a computing system facilitating DDPI in a network, in accordance with an aspect of the present application. A computing system **600** can include a set of processors **602**, a memory unit **604**, a NIC **606**, and a storage media **608**. Memory unit **604** can include a set of volatile memory devices (e.g., dual in-line memory module (DIMM)). Furthermore, computing system **600** may be coupled to a display device **612**, a keyboard **614**, a pointing device **616**, if needed, and a network interface card (NIC) **606**. Storage media **608** can store an operating system **618** and a DDPI system **620**. DDPI system **620** can be deployed in conjunction with an NMS running on computing system **600**.

(62) DDPI system **620** can include instructions, which when executed by computing system **600**, can cause computing system **600** to perform methods and/or processes described in this disclosure. Specifically, DDPI system **620** can include instructions for aggregating discovery information from the discovery packets received from the network devices of a network (aggregation process **622**). DDPI system **620** can also include instructions for determining how the network devices of the network are connected to each other (connectivity process **624**). Hence, DDPI system **620** can include instructions for determining the topology of the network, which indicates the hierarchy of the network devices in the network (connectivity process **624**).

(63) In addition, DDPI system **620** can include instructions for classifying the network devices into corresponding connectivity tiers (e.g., based on a bucketing algorithm) (classification process **626**). Moreover, DDPI system **620** can include instructions for allocating respective personas to the network devices in accordance with the connectivity tiers (allocation process **628**). DDPI system **620** can also include instructions for enhancing the determination of the connectivity tiers (e.g., by presenting an interface and receiving feedback from a user via the interface) (enhancement process **630**).

(64) Moreover, DDPI system **620** can also include instructions for configuring the network devices based on the configurations indicated by corresponding (e.g., by generating instruction packets comprising the configurations) (configuration process **632**). Furthermore, DDPI system **620** can include instructions for sending and receiving packets (e.g., receiving the discovery packets and sending the instruction packets) (communication process **634**). Data **636** can include any data that can facilitate the operations of DDPI system **620**. Data **636** can include, but is not limited to, discovery information, a set of device profiles, predetermined personas, and information indicating the allocation of connectivity tiers and personas.

(65) FIG. **7** illustrates an example of a non-transitory computer-readable storage medium that facilitates DDPI in a network, in accordance with an aspect of the present application. Computer-readable storage media **700** can comprise a plurality of instruction sets, when executed by processing resources, can perform functions or operations similar to processes **622-634** of DDPI system **620** of FIG. **6**. Specifically, storage media **700** can include aggregation instructions **702**; connectivity instructions **704**, classification instructions **706**; allocation instructions **708**; enhancement instructions **710**; configuration instructions **712**; and communication instructions **714**, which can correspond to processes **622-634**, respectively, of DDPI system **620** of FIG. **6**.

(66) The description herein is presented to enable any person skilled in the art to make and use the

invention, and is provided in the context of a particular application and its requirements. Various modifications to the disclosed examples will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other examples and applications without departing from the spirit and scope of the present invention. Thus, the present invention is not limited to the examples shown, but is to be accorded the widest scope consistent with the claims.

(67) One aspect of the present technology can provide an NMS that can provision and manage a network. During operation, the NMS can determine, for a respective network device in the network, connectivity information indicating how the network device is connected within the network and a device profile of the network device. The NMS can then determine a connectivity tier of the network device in the network based on the connectivity information. The connectivity tier can correspond to a placement of the network device in a hierarchy of network devices in the network. The NMS can select a device persona associated with the connectivity tier. Here, the device persona indicates specification and configurations for the network device. The NMS can send, to the network device, configuration information that configures the device persona on the network device.

(68) In a variation on this aspect, the NMS can determine the connectivity information and the device category by receiving an LLDP packet based on a logical connection with the network device and obtaining discovery information from the LLDP packet. The NMS can also determine the connectivity information based on the discovery information and determine the device profile based on identifying information associated with the network device in the LLDP packet.

(69) In a variation on this aspect, the NMS can determine the connectivity tier of the network device by determining a number of connections associated with the network device and determining an access tier as the connectivity tier of the network device if the number of connections is less than a threshold.

(70) In a further variation, if the number of connections is greater than the threshold, the NMS can determine whether the network device is connected to the access tier. If the network device is connected to the access tier, the NMS can determine an aggregation tier as the connectivity tier of the network device. On the other hand, if the network device is not connected to the access tier, the NMS can determine a core tier as the connectivity tier of the network device.

(71) In a further variation, the NMS can send configuration information to a respective network device at the access tier prior to sending configuration information to a network device at the aggregation tier.

(72) In a variation on this aspect, the NMS can validate the connectivity tier against the device profile of the network device to determine whether the connectivity tier is correctly allocated. The device profile can comprise vendor-specific information.

(73) In a variation on this aspect, the NMS can determine that the network device is connected to another network device at the same connectivity tier. The NMS can then select the device persona to indicate that the network device is in a device pair facilitating high availability to each other.

(74) In a variation on this aspect, the NMS can maintain, in a repository at the NMS, one or more predefined personas for a respective connectivity tier. The NMS can then select the persona for the network device from the repository based on the connectivity tier of the network device.

(75) In a variation on this aspect, the NMS can present, on a user interface of the NMS, the determination of the connectivity tier of the network device. The NMS can then receive, from a user via the user interface, an instruction to change the connectivity tier for the network device.

(76) The data structures and code described in this detailed description are typically stored on a computer-readable storage medium, which may be any device or medium that can store code and/or data for use by a computer system. The computer-readable storage medium includes, but is not limited to, volatile memory, non-volatile memory, magnetic and optical storage devices such as disks, magnetic tape, CDs (compact discs), DVDs (digital versatile discs or digital video discs), or other media capable of storing computer-readable media now known or later developed.

(77) The methods and processes described in the detailed description section can be embodied as code and/or data, which can be stored in a computer-readable storage medium as described above. When a computer system reads and executes the code and/or data stored on the computer-readable storage medium, the computer system performs the methods and processes embodied as data structures and code and stored within the computer-readable storage medium.

(78) The methods and processes described herein can be executed by and/or included in hardware logic blocks or apparatus. These logic blocks or apparatus may include, but are not limited to, an application-specific integrated circuit (ASIC) chip, a field-programmable gate array (FPGA), a dedicated or shared processor that executes a particular software logic block or a piece of code at a particular time, and/or other programmable-logic devices now known or later developed. When the hardware logic blocks or apparatus are activated, they perform the methods and processes included within them.

(79) The foregoing descriptions of examples of the present invention have been presented only for purposes of illustration and description. They are not intended to be exhaustive or to limit this disclosure. Accordingly, many modifications and variations will be apparent to practitioners skilled in the art. The scope of the present invention is defined by the appended claims.

Claims

1. A method comprising: determining, by a network management system for a network device in a network, connectivity information indicating how the network device is connected within the network and a device profile of the network device, wherein the network management system comprises a hardware processor and is to provision and manage the network, and wherein the connectivity information comprises a connectivity number representing how many connections the network device has to other network devices; determining, by the network management system, a connectivity tier of the network device in the network based on comparing the connectivity number to a threshold, wherein the connectivity tier corresponds to a placement of the network device in a hierarchy of network devices in the network; selecting, by the network management system, a device persona associated with the connectivity tier, wherein the device persona indicates a configuration for the network device; and sending, by the network management system to the network device, configuration information that configures the network device based on the device persona.
2. The method of claim 1, wherein the determining of the connectivity information comprises: receiving, by the network management system, a Link Layer Discovery Protocol (LLDP) packet based on a logical connection with the network device, wherein the logical connection indicates the network device being provisioned and managed by the network management system, obtaining discovery information from the LLDP packet, and determining the connectivity information based on the discovery information.
3. The method of claim 1, wherein the determining of the connectivity information for the network device comprises: exploring paths in the network from the network device to the other network devices, and computing the connectivity number based on identifying how many paths reach the other network devices from the network device.
4. The method of claim 1, further comprising: based on determining that the connectivity number is less than the threshold, determining an access tier as the connectivity tier of the network device.
5. The method of claim 4, further comprising: based on determining that the connectivity number is above the threshold, determining whether the network device is connected to the access tier; and based on determining that the network device is connected to the access tier, determining an aggregation tier as the connectivity tier of the network device.
6. The method of claim 1, further comprising validating the connectivity tier against the device profile of the network device to determine whether the connectivity tier is correctly allocated,

wherein the device profile comprises vendor-specific information for the network device.

7. The method of claim 1, further comprising: determining that the network device is connected to another network device at the same connectivity tier; and selecting the device persona to indicate that the network device is in a device pair facilitating high availability to each other.

8. The method of claim 5, further comprising: based on determining that the network device is not connected to the access tier, determining a core tier as the connectivity tier of the network device.

9. The method of claim 3, wherein the exploring of the paths comprises: traversing the paths in the network that are within a specified number of hops from the network device to the other network devices.

10. A non-transitory computer-readable storage medium storing instructions that when executed cause a network management system comprising a hardware processor to: determine, for a network device in a network, connectivity information indicating how the network device is connected within the network, wherein the network management system provisions and manages the network, and wherein the connectivity information comprises a connectivity number representing how many connections the network device has to other network devices; determine a connectivity tier of the network device in the network based on comparing the connectivity number to a threshold, wherein the connectivity tier corresponds to a placement of the network device in a hierarchy of network devices in the network; select a device persona associated with the connectivity tier, wherein the device persona indicates a configuration for the network device; and send, from the network management system to the network device, configuration information that configures the network device based on the device persona.

11. The non-transitory computer-readable storage medium of claim 10, wherein the determining of the connectivity information comprises: receiving a Link Layer Discovery Protocol (LLDP) packet based on a logical connection with the network device, wherein the logical connection indicates the network device being provisioned and managed by the network management system, obtaining discovery information from the LLDP packet, and determining the connectivity information based on the discovery information.

12. The non-transitory computer-readable storage medium of claim 10, wherein the determining of the connectivity information for the network device comprises: exploring paths in the network from the network device to the other network devices, and computing the connectivity number based on identifying how many paths reach the other network devices from the network device.

13. The non-transitory computer-readable storage medium of claim 12, wherein the exploring of the paths comprises: traversing the paths in the network that are within a specified number of hops from the network device to the other network devices.

14. The non-transitory computer-readable storage medium of claim 10, wherein the instructions when executed cause the network management system to: based on determining that the connectivity number is less than the threshold, determine an access tier as the connectivity tier of the network device.

15. The non-transitory computer-readable storage medium of claim 10, wherein the instructions when executed cause the network management system to validate the connectivity tier against a device profile of the network device to determine whether the connectivity tier is correctly allocated, wherein the device profile comprises information of device capabilities of the network device.

16. The non-transitory computer-readable storage medium of claim 10, wherein the instructions when executed cause the network management system to: determine that the network device is connected to another network device at the same connectivity tier; and select the device persona to indicate that the network device is in a device pair facilitating high availability to each other.

17. The non-transitory computer-readable storage medium of claim 14, wherein the instructions when executed cause the network management system to: based on determining that the connectivity number is above the threshold, determine whether the network device is connected to

the access tier; and based on determining that the network device is connected to the access tier, determine an aggregation tier as the connectivity tier of the network device.

18. The non-transitory computer-readable storage medium of claim 10, wherein the instructions when executed cause the network management system to: present, on a user interface of the network management system, the connectivity tier of the network device; and receive, via the user interface, an instruction to change the connectivity tier for the network device.

19. A computer system comprising: a processor; and a non-transitory machine-readable medium comprising instructions executable on the processor to: provision and manage a network comprising a plurality of network devices; determine, for a network device in the network, connectivity information indicating how the network device is connected within the network, wherein the connectivity information comprises a connectivity number representing how many connections the network device has to other network devices; determine a connectivity tier of the network device in the network based on comparing the connectivity number to a threshold, wherein the connectivity tier corresponds to a placement of the network device in a hierarchy of the plurality of network devices; select a device persona associated with the connectivity tier, wherein the device persona indicates a configuration for the network device; and send, to the network device, configuration information that configures the network device based on the device persona.

20. The computer system of claim 19, wherein the determining of the connectivity information for the network device comprises: exploring paths in the network from the network device to the other network devices, and computing the connectivity number based on identifying how many paths reach the other network devices from the network device.
